

revisit-rich circuit

90° FoV

ToF beams

2 m

The diagram shows a square arena with a side length of 2 m. A dashed blue circle is centered in the arena. A black triangle at the bottom center represents the robot's position. Two orange lines, labeled 'ToF beams', extend from the robot to the right and bottom-right edges of the arena. A dashed blue arc indicates a 90° field of view (FoV) centered on the robot. Several gray rectangular obstacles are scattered within the arena, mostly along the top and left sides of the dashed circle.

A stylized illustration of a grey fly. It has a segmented body, two large wings, and three legs. The fly has a green circular eye, a blue circular antenna base, and an orange circular antenna tip.

- Event camera (DVS, 256 px / 90°)
- 3-beam ToF rangefinder (0.1–2.83 m)
- IMU (2-axis accel. + pitch-rate gyro)

Grid cells (2D position code)

Topological map

Appearance-keyed loop closure

- Appearance-hash match (cosine ≥ 0.60)
- Geometric pose-gate (< 0.25 m, < 0.35 rad)
- Pose-graph relaxation